

FIT BUDDIES

AI FITNESS PLAN GENERATOR

Using Gemini Models

PROJECT REPORT
Generative AI • Python • Gemini API
Personalized Fitness Planning System

Project Overview

Fit Buddies is a smart fitness planning application that uses a Gemini model to transform user preferences into a structured weekly activity plan.

Project Type	AI / Generative AI Application
Primary Language	Python
AI Model	Gemini
Core Function	Personalized Fitness Plan Generation

Prepared as a college mini-project / prototype report

01 | ABSTRACT & OBJECTIVES

Abstract

Fit Buddies is an AI-powered fitness plan generator that creates a structured activity schedule from user-provided preferences. The system collects information such as fitness goal, experience level, available time, and equipment. A structured prompt is then sent to a Gemini model, which generates a readable weekly plan. The project demonstrates how generative AI can be integrated into a practical application while keeping the interface simple and user-friendly.

Problem Statement

Many users find it difficult to organize a consistent activity routine that fits their available time and resources. Predefined routines may not match every user's requirements. Fit Buddies addresses this by generating a plan dynamically from the user's stated preferences.

Objectives

- Generate personalized fitness activity plans using Gemini.
- Accept simple user preferences through a structured input form.
- Present a weekly schedule in an easy-to-read format.
- Demonstrate practical Gemini API integration using Python.
- Provide a foundation for future progress tracking and AI assistance.

Key Features

Feature	Description
AI Plan Generator	Creates a weekly plan from user preferences.
User Input	Goal, level, time, and equipment.
Plan Display	Organizes activities by day.
AI Assistant	Can be extended for fitness-related questions.
Progress Tracking	Can be extended to record completed activities.

02 | SYSTEM DESIGN

System Architecture

1	User Interface	Collects fitness requirements
2	Python Application	Validates input and builds the prompt
3	Gemini Model	Generates the personalized plan
4	Response Handler	Processes the generated response
5	Output Dashboard	Displays the weekly plan

Working Flow

- User enters fitness preferences.
- Python validates the submitted information.
- The application creates a structured Gemini prompt.
- Gemini generates a weekly fitness activity plan.
- The response is formatted and displayed to the user.
- The user can later regenerate or track the plan.

Technology Stack

Technology	Role
Python	Application logic and API integration
Gemini API / SDK	Generative AI plan creation
HTML + CSS	Optional web interface
JavaScript	Optional frontend interaction
Database	Optional profile and progress storage

03 | PYTHON + GEMINI IMPLEMENTATION

The prototype below demonstrates the core AI-generation logic. Replace the placeholder API key with a securely stored key when running the application.

Python Source Code

```
from google import genai

# Initialize Gemini client
client = genai.Client(api_key="YOUR_GEMINI_API_KEY")

goal = input("Fitness goal: ")
level = input("Experience level: ")
minutes = input("Time per day (minutes): ")
equipment = input("Equipment available: ")

prompt = f"""
You are Fit Buddies, an AI fitness planning assistant.

Create a simple weekly activity plan.

Goal: {goal}
Level: {level}
Available time: {minutes} minutes/day
Equipment: {equipment}

Return:
Day | Activity | Duration | Notes

Keep the plan practical, include recovery,
and avoid unsafe or extreme recommendations.
"""

response = client.models.generate_content(
    model="gemini-2.5-flash",
    contents=prompt
)

print("\n===== FIT BUDDIES =====")
print(response.text)
```

Code Explanation

- **Client initialization:** Connects the Python application to Gemini.
- **User inputs:** Collects the goal, experience, time, and equipment.
- **Prompt construction:** Combines the inputs into clear instructions.
- **Model request:** Sends the prompt to the selected Gemini model.
- **Response:** Prints the generated plan for the user.

04 | SAMPLE INPUT & OUTPUT

Sample Input

Fitness goal: General fitness
Experience level: Beginner
Time per day: 30 minutes
Equipment available: No equipment

Sample Gemini Output

DAY	ACTIVITY	DURATION	NOTES
Monday	Light cardio + mobility	20–30 min	Comfortable pace
Tuesday	Basic bodyweight activity	20–30 min	Controlled movement
Wednesday	Easy walk / recovery	15–30 min	Keep activity light
Thursday	Cardio + mobility	20–30 min	Adjust to comfort
Friday	Basic strength activity	20–30 min	Rest as needed
Saturday	Enjoyable physical activity	20–40 min	Choose a safe activity
Sunday	Rest / recovery	—	Recovery day

Console Output Example

```
===== FIT BUDDIES =====  
Personalized Weekly Fitness Plan  
  
Monday   : Light cardio + mobility  
Tuesday  : Basic bodyweight activity  
Wednesday : Easy walk / recovery  
Thursday : Cardio + mobility  
Friday   : Basic strength activity  
Saturday : Enjoyable physical activity  
Sunday   : Rest / recovery
```

05 | BENEFITS, SAFETY & CONCLUSION

Advantages

- Personalized output based on the user's stated requirements.
- Simple and scalable generative-AI architecture.
- Can be integrated into a website or mobile application.
- Easy to regenerate when preferences change.
- Provides a strong base for future progress-tracking features.

Safety & Responsible Use

Fit Buddies is a planning prototype, not a medical diagnostic system. AI-generated suggestions should be treated as general information. The application should avoid extreme exercise, dangerous challenges, and restrictive eating recommendations. Users should stop an activity if they feel unwell and seek appropriate professional advice when needed.

Future Enhancements

- User login and personalized profiles.
- Workout completion tracking and progress charts.
- AI-based plan adjustments using previous activity.
- Tamil and multilingual support.
- Voice-based fitness assistant.
- Wearable-device integration.
- Calendar reminders and notifications.

Conclusion

Fit Buddies demonstrates a practical use of Gemini generative AI for personalized fitness planning. The system accepts user requirements, constructs a structured prompt, generates a weekly plan, and presents the result in a clear format. With additional features such as authentication, progress tracking, multilingual support, and wearable integration, the prototype can be extended into a complete AI-powered fitness application.

PROJECT FLOW

USER INPUT → PYTHON → GEMINI AI → PERSONALIZED PLAN → OUTPUT

Implementation note: Gemini SDK/API names and model availability may change. Use the current Gemini documentation when running the code.